

INTERNATIONAL STUDENT PORTAL
TEAM: SMO2

Analysis and design report

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1. Introduction

Studying abroad is an exciting yet challenging experience. For international students, the journey involves not only traveling but also navigating complex administrative processes that can be time-consuming and overwhelming. From applying for admission to arranging accommodations and completing legal formalities, the current approach often requires considerable effort and involves multiple platforms, leading to confusion and delays.

Our project aims to simplify and streamline the administrative processes for international students, particularly those enrolling as degree-seeking or exchange students. By creating a centralized application/portal, we envision replacing the fragmented systems currently in use. This platform will facilitate seamless collaboration among all stakeholders, including students, the international office, and home institutions.

The project encompasses the entire administrative workflow, covering key aspects such as:

- Nomination
- Application
- Issuance of acceptance letters
- Registration at Thomas More
- Option to apply for a buddy
- Creation of the Online Learning Agreement (OLA)
- Access to informative webinars
- Assistance with housing searches

By consolidating these functions into a single, user-friendly platform, we aim to reduce confusion, minimize administrative bottlenecks, and alleviate the workload on international coordinators. This, in turn, will create a stress-free and efficient process, ensuring that students arrive at Thomas More well-prepared and with peace of mind.

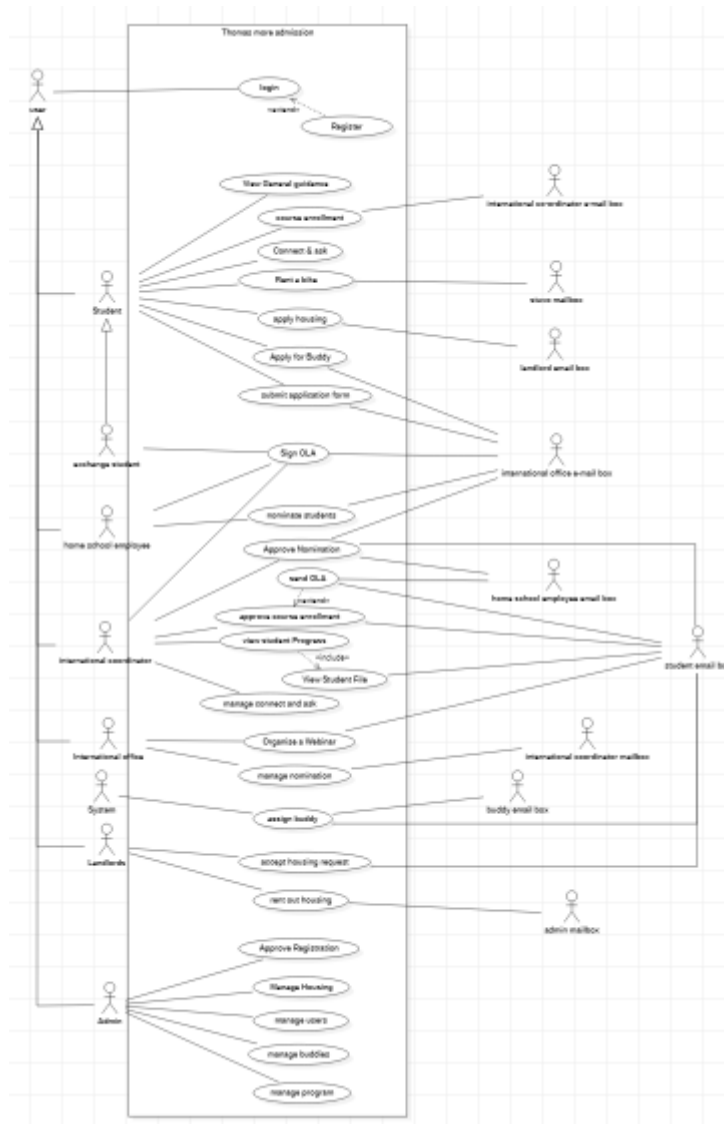
2. Requirements analysis

In this part, we look at the functional and non functional requirements for the project, for this we made a use case diagram showing all users (actors), both passive and active, as well as all the actions the system will perform.

After which we look at each of those actions in detail, describing the function, prerequisites, normal flow and alternative flows of each action)

2.1. Functional requirements

2.1.1. Use Case Diagram



2.1.2. Use Case Descriptions

2.1.2.1. Login

Functionality: As a user, I can log in to the system.

Normal Flow:

System asks actor to input their account details. Actor enters email address and password. System shows homepage.

Alternatives:

Wrong password: System shows error message. Actor can try again.

Email address doesn't exist: System shows error message. Actor chooses to register. System starts UC Register.

Reset password: Actor chooses to reset password and fills in email address. System sends email with new activation link.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=57-179&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=57%3A179>

2.1.2.2. Register

Functionality: As a user, I can register myself.

Normal Flow:

System asks the actor to input their name, email, role, password and option to upload files for their role confirmation. Actor fills in personal data. System sends these details to the admin to approve registration and displays a message to wait for registration approval.

Alternatives:

Email address already exists: System asks for a new email address.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=43-21&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=43%3A21>

2.1.2.3. View General Guidance

Functionality: As a student, I can see general information.

Precondition: Actor must be logged in.

Normal Flow:

System displays links to all relevant sections of the application, as well as information about all the general procedures including opening a bank account, registering at Town Hall, etc. inside of the FAQ.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=61-189&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=61%3A189>

2.1.2.4. Course Enrollment

Functionality: As a student, I can enroll in courses.

Precondition: Actor must be logged in.

Normal Flow:

System shows a list of the available courses (subject, course Id and credit values). Actor selects desired courses and presses view schedule. System shows the schedule for the given courses and asks for the confirmation. Actor confirms his choice. System saves Actor's enrollment choice, updates their profile and sends a notification to international coordinator.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-638&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=20%3A3>

2.1.2.5. Connect & Ask

Functionality: As a student, I can chat with other incoming students.

Precondition: Actor must be logged in.

Normal Flow:

System shows a list of discussion forums. Actor selects a forum. System shows the requested forum board and allows the actor to send messages.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=747-1816&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=62%3A242>

2.1.2.6. Rent a Bike

Functionality: As a student, I can rent a bike.

Precondition: Actor must be logged in.

Normal Flow:

System Shows the actor a bike rental form. Actor fills in bike rental form. System sends the form to stuvo's mailbox.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=62-287&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=62%3A287>

2.1.2.7. Apply Housing

Functionality: As a student, I can apply for housing.

Precondition: Actor must be logged in.

Normal Flow:

System shows a list of available dorm rooms with their rental price, available date and location. User applies for their desired dorm. System sends request to landlord's mailbox.

Alternative:

No available dorm room: System displays that all the dorm rooms are booked.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=62-374&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=62%3A374>

2.1.2.8. Apply for Buddy

Functionality: As a student, I can apply for a buddy.

Precondition: Actor must be logged in.

Normal Flow:

System requests actor to fill out personal details (name, email, address, country) and interests. Actor chooses preferences and fills form. System sends the information to international office mailbox.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=62-561&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=62%3A561>

2.1.2.9. Submit Application Form

Functionality: As a student, I can submit application form for admission.

Precondition: Actor must be logged in.

Normal Flow:

System shows a form for the actor to fill in name, email, desired program, country, previous education and upload educational document . Actor fills in personal details and program. System saves data and sends the email to international office email box.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=62-619&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=62%3A619>

2.1.2.10. Sign OLA

Functionality: As an exchange student, international coordinator and home school employee, I can digitally sign OLA.

Precondition: Actor must be logged in and OLA must be already created by system.

Normal Flow:

System displays online learning agreement document showing all the courses and credits student has enrolled for. Actor signs document digitally by presses the sign button. System shows an option download the document.

Alternative:

OLA not signed: System sends notification to actor and reminds them to sign within given time.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-13054&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=63%3A62>

2.1.2.11. Nominate Students

Functionality: As a home school employee, I can nominate students for admission.

Precondition: Actor must be logged in.

Normal Flow:

System requests details of the candidate (name, email, address, campus, desired course, country of origin, phone number, home school, previous education). Actor fills in candidate's data and desired course. System saves nominated student data and notifies international office.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=72-33&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=72%3A33>

2.1.2.12. Approve Nomination

Functionality: As an International coordinator, I can approve/disapprove nominations.

Precondition: Actor must be logged in.

Normal Flow:

System displays nominated student name, email, address, home school, desired course, previous qualifications, attached documents and available space in the desired course. Actor approves nomination. System sends email to home school employee and nominated student mailbox.

Alternative:

Not enough space: System shows no more available space. Actor rejects nomination. System notifies home school employee and nominated student.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-3357&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=31%3A27>

2.1.2.13. Send OLA

Functionality: As a system, I can send OLA.

Precondition: International Coordinator must approve the course enrollment for exchange student.

Normal Flow:

System creates OLA document including selected courses and credits and sends it to home school employee and student email box.

2.1.2.14. Approve Course Enrollment

Functionality: As an international coordinator, I can approve student's course enrollment.

Precondition: Actor must be logged in and student must have submitted courses.

Normal Flow:

System displays a list with name, email, program and status of the student enrollment. Actor selects one of the students from the list. System displays the courses selected by the student showing each course name, credits, course code, study year and phase in which that course is taught. Actor selects the option to check schedule of the student. System displays the timetable of the student showing class time for each course. Actor approves the course enrollment. System notifies the student by sending email to student email box.

Alternative:

Course enrollment disapproved: System notifies the student by sending email to student email box.

Is exchange student: if the course enrollment is approved for the exchange student, system creates OLA and its auto signed for international coordinator and system switches to the user case send OLA.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-3892&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=42%3A84>

2.1.2.15. View Student Progress

Functionality: As an international coordinator, I can view incoming student list and see the status of their admission.

Precondition: Actor must be logged in.

Normal Flow:

System displays a list of incoming student names, email, program and status of file. Actor selects one of the students to view the file. System switches to use case view student file

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=906-1913&t=fflC0hlnFm6GkwLg-0&scaling=min-zoom&content-scaling=fixed&page-id=217%3A813>

2.1.2.16. View Students File

Functionality: As an international coordinator, I can view incoming student files and check admission steps they have completed.

Precondition: Actor must be logged in.

Normal Flow:

System displays the student details (campus, address, home school, previous education, enrollment date, program) and the steps that student has completed regarding admission. Actor views the completed and uncompleted steps and sends remarks to the student if needed.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=48-330&p=f&t=aBivhwmm0cgJtrMp-0&scaling=min-zoom&content-scaling=fixed&page-id=48%3A330>

2.1.2.17. Manage Connect and Ask

Functionality: As an international coordinator, I can add discussion forums in the connect and ask.

Precondition: Actor must be logged in.

Normal Flow:

System displays all the discussion forum and option to add new forum. Actor add new forum by giving it a name and add students in it. System saves the forum. Actor can also edit an existing forum.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=819-2789&t=aBivhwmm0cgJtrMp-0&scaling=min-zoom&content-scaling=fixed&page-id=819%3A2788&starting-point-node-id=819%3A2789>

2.1.2.18. *Organize a Webinar*

Functionality: As an international office employee, I can organize a webinar.

Precondition: Actor must be logged in.

Normal Flow:

System requests actor to input webinar information and selects students for it to be sent to. Actor inputs information and submits. System shows success screen and sends notification to students.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-5410&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=19%3A2>

2.1.2.19. *Manage nominations*

Functionality: As an international office, I can manage nominations.

Precondition: actor must be logged in.

Normal flow:

System displays a list of nominated student info (name, number, program). Actor selects a student, system shows a more detailed description of the student, actor presses submit, system sends the nomination to a relative international coordinator.

Alternative:

Actor rejects nomination: system sends a mail to the homeschool and nominated student's mailbox containing a rejection letter.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=68-473&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=68%3A473>

2.1.2.20. *Assign buddy*

Functionality: As a system, I can assign a buddy.

Precondition: actor must be logged in

Normal flow: System matches the interest of buddies and incoming students that are in the system and assigns a buddy to an incoming student. System notifies the students when it has assigned.

2.1.2.21. *Accept housing request*

Functionality: As a landlord I can accept or decline housing request.

Precondition: Actor must be logged in.

Normal Flow: System shows the housing request made by students. Actor views the request. System shows the requested dorm with desired rental period. Actor can chat with the student to discuss about rental terms.

Alternative:

Room is already taken: Actor declines the request

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-6608&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=68%3A611>

2.1.2.22. Rent out housing

Functionality: As a landlord I can rent out housing

Precondition: Actor must be logged in.

Normal Flow:

System shows the form to fill in housing information including images, name, address, room number, price, renting period and amenities available. Actor fills in the form. System saves the information.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-6951&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=68%3A749>

2.1.2.23. Approve Registration

Functionality: As an admin, I can approve user registration.

Precondition: Actor must be logged in.

Normal Flow:

System displays a list of users who applied for registration showing their name, email, role and status of their account. Actor selects a user from the list. System displays option to reject or approve the registration. Actor approves the registration. System sends the email to user notifying that registration has been approved.

Alternative:

Registration rejected: If an actor rejects the registration, System sends an email to user notifying that registration is rejected.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-7452&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=719%3A3806>

2.1.2.24. Manage housing

Functionality: As an admin, I can manage (create, read, update and delete) housing.

Precondition: Actor must be logged in.

Normal Flow:

System displays list of the landlord names. Actor can select desired landlord from the list to view, update or delete his details and the dorm rooms he has provided in the system for the rent out. System updates the information (details of the landlord) accordingly.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-7987&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=52%3A13>

2.1.2.25. Manage User

Functionality: As an admin, I can manage (create, read, update and delete) users.

Precondition: Actor must be logged in.

Normal Flow:

System displays a list of the registered users showing their names, email, role and an option to create a new user. Actor can select desired user from the list to view, update or delete his details (name, address, email, roll etc). System updates the information accordingly.

Alternative:

Create new user: If an actor chooses to create a new user, system displays a form to fill in name, campus, email and role of the user. Actor fills in the form. System saves these details and adds the user in the system and sends the email to user notifying account creation.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=719-694&p=f&t=aBivhwmm0cgJtrMp-0&scaling=min-zoom&content-scaling=fixed&page-id=719%3A694>

2.1.2.26. *Manage Buddies*

Functionality: As an admin, I can manage (create, read, update and delete) buddies.

Precondition: Actor must be logged in.

Normal Flow:

System displays a list of the buddy names and an option to create a new buddy. Actor can select desired buddy from the list to view, update or delete his details (name, address, phone number, email, course enrolled, r-number. System updates the information (details of the buddy) accordingly.

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-8969&p=f&t=vATi3aBMPQ4YZ8qg-0&scaling=min-zoom&content-scaling=fixed&page-id=53%3A241>

2.1.2.27. *Manage Program*

Functionality: As an admin, I can manage (create, read, update and delete) courses.

Precondition: Actor must be logged in.

Normal Flow:

System displays list of the program names with their campus and language they are taught and an option to create new program. Actor can select desired program from the list to view, update or delete its details (name, subjects included, study points, etc). System updates the information (details of the course) accordingly.

Alternative:

Create new program: If an actor chooses to create a new program. System displays a form to fill in name, program coordinator, campus, duration, language and available seats for the program. Actor fills in the form. System saves these details and adds the program name in the list,

Figma prototype: <https://www.figma.com/proto/Hxd5byOUvf0tLz0T5ztmL/Wireframing?node-id=779-11347&p=f&t=aBivhwmm0cgJtrMp-0&scaling=min-zoom&content-scaling=fixed&page-id=67%3A207>

2.2. Non-functional requirements

File uploads: Allow the user to upload files in different formats (docx, pdf and images).

Reliability: The system should ensure high availability of 99.9% uptime, with minimal downtime to handle the constant influx of student and administrative data.

Scalability: As the number of international students increases, the system should scale efficiently to support additional users and transactions. It should support 100 plus concurrent users at beginning.

Performance: The application should offer fast response times, even under peak load conditions, ensuring smooth user experiences. Page Load should be less than 3 second.

Security: All sensitive student and institutional data must be securely stored and transmitted, adhering to GDPR and other relevant data protection regulations.

Accessibility: The portal should comply with accessibility standards to accommodate users with disabilities.

Authentication: Multi factor authentication for all users and user role to be authenticated properly.

Maintainability: The system should be modular and easy to update or enhance as new features are required.

Integration: Seamless integration with existing systems (e.g., email, webinar tools, housing databases) is essential for operational efficiency.

Log File: Track all users actions to detect any irregularities.

Multi Language Support: As the application is for international student so primary language must be English but application must have Dutch as secondary language.

University Regulations: Application must align with academic policies.

3. Priority by functionality

Use Cases	Must Have	Should have	Could have	Won't have
Login	x			
Register	x			
View general guidance		x		
Courser enrolment	x			
Connect and Ask			x	
Rent a bike		x		
Apply housing			x	
Apply for buddy		x		
Submit application form	x			

Use Cases	Must Have	Should have	Could have	Won't have
Sign OLA		x		
Nominate students	x			
Approve nominations	x			
Send OLA		x		
Approve course enrolment	x			
View student progress	x			
View student file	x			
Manage connect and ask			x	
Organize a webinar			x	
Manage Nomination	x			
Assign buddy		x		
Accept housing request			x	
Rent out housing			x	
Approve registrations	x			
Manage housing			x	
Manage users		x		
Manage buddies		x		
Manage program	x			

4. Sources

All work is original and nothing is taken from internet or any other resource of information. ChatGPT has been used to correct grammatical errors.

5. Conclusion

In conclusion, this project seeks to revolutionize the experience of international students by centralizing and streamlining their administrative journey. By integrating various essential processes—ranging from

admission and course enrollment to housing and mentorship—into a unified platform, the system will significantly reduce complexity and improve efficiency for all stakeholders. The resulting reduction in administrative workload, combined with a user-friendly interface and secure operations, will create a more welcoming and supportive environment for international students. Ultimately, this initiative will not only enhance student satisfaction but also uphold the institution's commitment to fostering an inclusive and globally connected academic community.